



User Manual

Apni Chhat Apna Ghar — Construction Validation App

Government of Punjab · The Urban Unit

This guide covers both app roles:

1. Field Engineer
2. House Owner (Homeowner)

Demo project: ACAG-1 · Flutter + Supabase

1. Getting Started

1.1 Open the app

- Launch ACAG on your phone.
- Wait for the splash screen.
- You will see the Login screen.

1.2 Demo accounts (testing / FYP demo)

Role	Email	Password
Field Engineer	shoaibkhilji141@gmail.com	12345678
House Owner	ali.raza.owner@gmail.com	12345678

Demo project linked to both accounts: ACAG-1
Owner: Ali Raza · Engineer: Shoaib Khilji

1.3 Login

- Enter email and password.
- Tap Login.
- The app opens the correct home screen for y

1.4 Common UI elements

- Top bar: title / branding and notification bell (
- Back arrow: on screens opened from another
- Pull down to refresh: reloads latest data on m
- Bottom navigation: switch main sections.

2. Field Engineer Guide

2.1 Bottom navigation

Tab	Purpose
Dashboard	Overview, stats, recent notifications
Projects	Assigned house projects list
Camera (center)	Quick site photo !' visit logged
Reports	Module completion certificates
Profile	Profile, documents, settings, logout

2.2 Dashboard

- See assigned projects summary.
- Open notifications from the bell icon.
- Use shortcuts to Projects or key actions.

2.3 Assigned Projects

- Open Projects and tap a project (e.g. ACAG-1).
- Review Project ID, address, district/tehsil, progress %, owner, n

2.4 Construction Modules (01–05)

Module	What you do
01 Planning & Elevation	Plot, rooms, floor plans, elevation
02 Structure	Stories, soil, foundation, frame type
03 Materials	Material estimation
04 Construction Tracking	Stages, photos, quality assessment
05 Handover	Handover, HSE, completion certificate

Note: Completing a module can notify the owner. Visit/completion time.

2.5 Site photos & engineer visits

Option A — Camera button (bottom center)

- Tap the camera FAB !' Take Photo or Gallery.
- Photo is saved to the assigned project.
- App logs an engineer visit (visited now, next visit "H +7 days).

Option B — Module / project upload

- Upload progress photos with captions in Module 04 flows.

2.6 Reports

- Open Reports tab.
- View module completion certificates for finished modules.
- Tap a report to open the certificate view.

2.7 Notifications (Engineer)

- Open the bell icon or Profile !' Notifications.

You may receive:

- Owner complaint submitted
- Owner feedback / rating submitted
- Module / visit related updates

2.8 Profile (Engineer)

- View/edit name, phone, location and profile picture.
- Documents: tap CNIC (etc.) to open picture viewer.
- Logout from the bottom of the screen.

3. House Owner Guide

3.1 Bottom navigation

Tab	Purpose
Dashboard	Progress, shortcuts, recent notifications
My Project	Full house / project details
Reports	Module reports + documents
Profile	Profile, documents, settings, logout

3.2 Dashboard

- Overall construction progress %.
- Quick links: Project, Notifications, Visits.
- Shortcuts: Feedback, Complaints, Documents.
- Recent notifications list.

3.3 My Project (House information)

You can view:

- Project / House ID, address, city / district / tehsil
- Plot size, covered area, number of floors
- Start date, expected completion, progress %
- Module status (read-only), images & materials

Track Construction actions:

- Construction Progress · Site Photos · Engineer Visits
- Documents · Complaints / Issues · Feedback & Rating

3.4 Construction Progress

- Open Construction Progress from My Project.
- See the stage timeline from site prep to completion.
- Tap a stage for status, dates, remarks, and photos.

Stages show Completed after the engineer records them in Module 04; otherwise Pending.

3.5 Engineer Visits

- Open Engineer Visits from Dashboard or My Project.
- See assigned engineer, date/time, purpose, remarks, next visit.
- Visits are logged when the engineer uploads a site photo.

3.6 Site Photographs

- Browse photos uploaded by the engineer.
- Each item can show caption, date, and uploader name.
- Tap to view larger.

3.7 Documents

Open from: Reports !' Documents, Profile !' Documents, or Dashboard shortcut.

Document types:

- CNIC · Property · Construction plan · NOC
- Loan/grant · Engineer reports · Completion certificate

How to view

- Tap a document row (e.g. CNIC).
- Viewer opens with image(s); swipe Front/Back for CNIC.
- Pinch-to-zoom where supported; back arrow returns to list.

3.8 Complaints / Issues

- Open Complaints !' New Complaint.
- Fill category, priority, description, location, optional photos.
- Submit — status starts as Submitted.
- Workflow: Submitted !' Under Review !' In Progress !' Resolved.
- Your assigned engineer receives an in-app notification.

3.9 Feedback & Rating

- Select 1–5 stars, optional category and comments.
- Submit Feedback — engineer is notified in-app.

3.10 Notifications (Owner)

- Open the bell icon for visit/module/project alerts.

Owner alerts are in-app only (no external push notifications).

3.11 Profile (Owner)

- Name, CNIC, phone, email, address, photo, account status.
- Documents section with the same viewer.
- Edit Profile, Notifications, Logout.

4. Notifications — Quick Reference

Event	Who gets notified
Owner submits complaint	Assigned Engineer
Owner submits feedback	Assigned Engineer
Engineer completes a module	Project parties (incl. Owner)
Site photo !' visit logged	Project parties (incl. Owner)

5. Tips for Smooth Use

- Internet is required — data loads from the cloud.
- First open may be slower; later opens use cache.
- Pull-to-refresh if data looks outdated.
- Engineer should upload photos regularly for visits & evidence.
- Owner is read-only on construction data (except complaints + f

6. Troubleshooting

Problem	What to try
Cannot login	Check email/password; use demo accounts
Empty project/stages	Use correct role; demo project is ACAG-1
No notifications	Bell !' refresh; engineer must be assigned
Document has no image	Refresh; ensure docs seeded for project
App feels slow	Use Wi-Fi; refresh once; avoid huge photo loops
Back button missing	Use system back, or open via shortcut

7. Role Summary

Engineer can

- Manage assigned projects and complete Modules 01–05
- Upload site photos (auto visit log)
- View reports/certificates and project documents
- Receive owner complaints & feedback

Owner can

- View profile, house info, progress, photos, visits, documents
- Submit complaints & feedback
- Read in-app notifications

Owner cannot

- Edit project meta (start date, district, covered area)
- Complete engineer modules or change visit schedule directly

8. Support

For FYP / demo support, contact your project team or Urban Unit c

App: ACAG — Apni Chhat Apna Ghar

Stack: Flutter + Supabase

